



Modelling Financial Markets

Chapter 11: Continuous-Time Models



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Acronyms used in this material

- ▣ ACF: AutoCorrelation Function
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CBOE: Chicago Board Options Exchange
- ▣ CI: Confidence Interval
- ▣ CIR: Cox–Ingersoll–Ross (model)
- ▣ CLT: Central Limit Theorem
- ▣ DTB3: 3-Month Treasury Bill: secondary market rate (FRED series code)
- ▣ FRED: Federal Reserve Economic Data
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GBM: Geometric Brownian Motion
- ▣ JB: Jarque–Bera (test)
- ▣ LR: Likelihood Ratio (test)
- ▣ MLE: Maximum Likelihood Estimation
- ▣ OLS: Ordinary Least Squares
- ▣ OU: Ornstein–Uhlenbeck (process)
- ▣ QQ: Quantile–Quantile (plot)
- ▣ SDE: Stochastic Differential Equation
- ▣ SVJ: Stochastic Volatility with Jumps (model)
- ▣ US: United States
- ▣ VIX: CBOE Volatility Index
- ▣ XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

Outline

Motivation

From Random Walk to Brownian Motion

Quadratic Variation and Itô Calculus

Stochastic Differential Equations and Simulation

Geometric Brownian Motion

Mean Reversion: the Ornstein–Uhlenbeck Process

Jumps: the Merton Model

Stochastic Volatility: the Heston Model

Models against the Data

References

Why Continuous Time?

The question of this chapter

- Prices are recorded daily or every few milliseconds, but trading never really stops
- Can we describe prices with a model that works at **every** time scale at once?
- Option pricing (Chapter 12), interest-rate models and risk management are written in this language

What we do in this chapter

- Build Brownian motion and Itô calculus through simulation
- Simulate stochastic differential equations and measure the error of the schemes
- Estimate four models on data: geometric Brownian motion, Ornstein–Uhlenbeck, Merton jumps, Heston volatility
- Confront every model with the stylised facts of Chapter 1

Robert Brown, 1827: Pollen Grains in Water



Robert Brown (1773–1858), botanist

Photo: Maull & Polyblank (1855); public domain;
Wikimedia Commons

- The Scottish botanist Robert Brown watched particles from pollen grains under a microscope
 - ▶ They moved without stop, in an irregular zig-zag, even in dead or mineral material
- The cause, collisions with water molecules, was explained only in 1905 [Einstein \(1905\)](#)
- The mathematical object behind it is called **Brownian motion**
- Finance reached the same object five years before physics [Bachelier \(1900\)](#)

Bachelier 1900 and Einstein 1905



Louis Bachelier (1870–1946)

Photo: unknown author; public domain; Wikimedia Commons

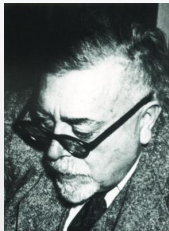


Albert Einstein in 1905

Photo: Lucien Chavan / ETH Zürich (1905); public domain; Wikimedia Commons

- Bachelier (1900): prices of French government bonds (rentes) and their options follow a random walk in continuous time
 - ▶ The spread of the price grows with $\sqrt{t}$
- Einstein (1905): the same law for particles in a fluid
 - ▶ The mean squared displacement grows linearly in time
- Bachelier was forgotten for fifty years; Samuelson rediscovered him in the 1950s

Wiener and Itô: the Mathematics



Norbert Wiener (1894–1964)

Photo: Konrad Jacobs; CC BY-SA
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Kiyosi Itô (1915–2008)

Photo: Konrad Jacobs (1970); CC
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- Wiener (1923): Brownian motion as a probability measure on the space of continuous paths
 - ▶ This is why it is also called the **Wiener process**
- Itô (1944): an integral with respect to Brownian motion and the chain rule for it
 - ▶ Itô's lemma is the main working tool of this chapter
- Textbook treatment for finance: Franke, Härdle & Hafner (2019), Part I



From Samuelson to Black, Scholes and Merton



Paul A. Samuelson (1915–2009)

Photo: Bernard Gotfryd
(1970–1975); public domain;
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Robert C. Merton, Nobel Prize
1997

Photo: Digarnick (2006); public
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- Osborne (1959): log prices, not prices, behave like Brownian motion
- Samuelson (1965): geometric Brownian motion, prices can never become negative
- Black & Scholes (1973) and Merton (1973): an option can be replicated in continuous time
- Merton (1976) adds jumps; Heston (1993) adds random volatility

Data Used in This Chapter

Series	Type	Period	Days
S&P 500	US equity index	1990–2026	9 240
BET	BVB equity index	2000–2026	6 670
Bitcoin	crypto-asset, 7 days a week	2014–2026	4 384
VIX	CBOE implied volatility index of the S&P 500	1990–2026	daily
DTB3	3-month US Treasury bill rate (FRED)	1954–2026	daily

- Daily closing values up to 18 September 2026; daily log returns $r_t = \ln P_t - \ln P_{t-1}$
- Each series on its own calendar; equity indices on weekdays with an unchanged close removed

Conventions

- ▣ Time t is measured in **years**; one trading day is $\Delta t = 1/252$ (Bitcoin: $1/365$)
 - ▶ All drifts and volatilities are annualised: $\mu = 10\%$ means 10% per year
- ▣ W_t : standard Brownian motion; dW_t : its increment over an infinitesimal interval dt
- ▣ S_t : price; $X_t = \ln S_t$: log price; v_t : instantaneous variance; σ : volatility
- ▣ **Stochastic process**: a family of random variables $\{X_t\}$ indexed by time
 - ▶ A **path** (trajectory) is one realisation $t \mapsto X_t(\omega)$
- ▣ Information up to t : the **filtration** $\mathcal{F}_t$, everything observed until time t

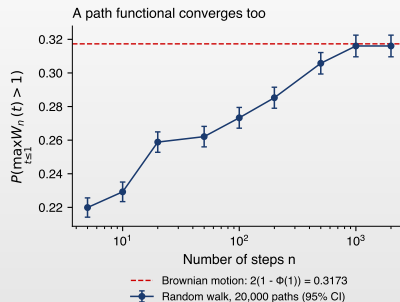
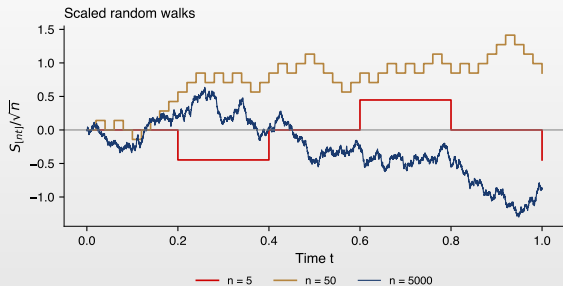
The Random Walk

- Steps $\xi_1, \xi_2, \dots$ independent, $P(\xi_i = +1) = P(\xi_i = -1) = 1/2$; position $S_k = \xi_1 + \dots + \xi_k$
 - ▶ $\mathbb{E}[S_k] = 0$, $\text{Var}(S_k) = k$: the spread grows like $\sqrt{k}$
- **Scaled random walk** on $[0, 1]$ with n steps: $W_n(t) = S_{\lfloor nt \rfloor} / \sqrt{n}$
 - ▶ Time step $1/n$, space step $1/\sqrt{n}$: the only scaling with a non-trivial limit
 - ▶ $\text{Var}(W_n(t)) \approx t$ for every n
- By the central limit theorem (CLT): $W_n(t) \rightarrow N(0, t)$ in distribution for each fixed t

Donsker's Invariance Principle

- The CLT concerns one time point; Donsker's theorem concerns the **whole path**
 - ▶ $W_n \Rightarrow W$: the scaled random walk converges in distribution, as a random function, to Brownian motion
Billingsley (1999)
- "Invariance": the limit does not depend on the distribution of the steps
 - ▶ Any i.i.d. steps with mean 0 and variance 1 give the same Brownian motion
- Consequence: every continuous functional of the path converges too
 - ▶ Example: the maximum; by the reflection principle $P(\max_{t \leq 1} W_t > a) = 2P(W_1 > a)$
- For finance: many small independent trades add up to Brownian-like price paths

Random Walks Converge to Brownian Motion



□ Left: one scaled random walk for $n = 5, 50, 5000$ steps; right: probability that the path exceeds 1 before $t = 1$

Reading the Donsker Chart

- With 5 steps the walk is a staircase; with 5000 steps it already looks like a Brownian path
 - ▶ The vertical scale does not blow up: the $\sqrt{n}$ scaling keeps $\text{Var}(W_n(1)) = 1$
- $P(\max W_n > 1)$ rises from 0.220 ($n = 5$) to 0.316 ($n = 2000$)
 - ▶ The Brownian value is $2(1 - \Phi(1)) = 0.3173$, with Φ the standard Normal distribution function
- A coarse grid **misses** the maximum between observations: daily closes understate the intraday range
- The convergence is slow, of order $1/\sqrt{n}$ for the maximum

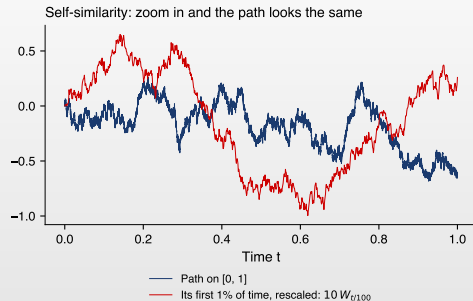
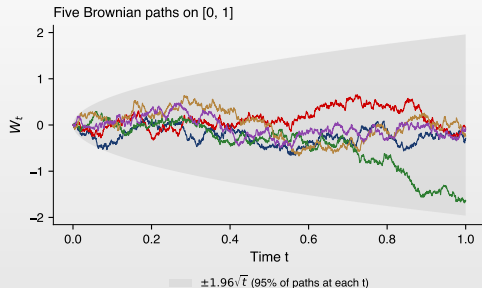
Definition: Brownian Motion

- A stochastic process $W = \{W_t\}_{t \geq 0}$ is a **standard Brownian motion** if
- (i) $W_0 = 0$
- (ii) independent increments: $W_{t_2} - W_{t_1}, W_{t_3} - W_{t_2}, \dots$ are independent for $t_1 < t_2 < t_3 < \dots$
- (iii) Normal increments: $W_t - W_s \sim N(0, t - s)$ for $s < t$
- (iv) continuous paths: $t \mapsto W_t$ is continuous
- Existence is a theorem [Wiener \(1923\)](#); simulation uses (ii) and (iii) on a grid: $W_{t+\Delta t} = W_t + \sqrt{\Delta t} Z$, $Z \sim N(0, 1)$

Properties of Brownian Motion

- Moments: $\mathbb{E}[W_t] = 0$, $\text{Var}(W_t) = t$, $\text{Cov}(W_s, W_t) = \min(s, t)$
- **Martingale**: $\mathbb{E}[W_t | \mathcal{F}_s] = W_s$ for $s < t$, the best forecast is today's value
 - ▶ The continuous-time version of the efficient-market random walk of Chapter 2
- **Self-similarity**: $\{c^{-1/2}W_{ct}\}$ is again a standard Brownian motion for any $c > 0$
- Paths are continuous but **nowhere differentiable**
 - ▶ $(W_{t+h} - W_t)/h$ has standard deviation $1/\sqrt{h} \rightarrow \infty$: ordinary calculus fails

Brownian Paths



- Left: five paths with the band $\pm 1.96\sqrt{t}$; right: a path and its first 1% of time, stretched 100 times in time and 10 times in space

Reading the Brownian Paths

- The band widens like $\sqrt{t}$, not like t : uncertainty over one year is $\sqrt{252} \approx 15.9$ times the daily one
 - ▶ The square-root-of-time rule used for VaR in Chapter 7 comes from here
- Paths wander far from zero and cross it again: no mean reversion
- Zoomed and rescaled, a piece of the path looks like the whole path
 - ▶ No time scale is special: the same model for minutes and for years
- Real markets break self-similarity: fat tails and clustering depend on the horizon (Chapter 1)

Recap: Brownian Motion

- Brownian motion is the limit of scaled random walks, whatever the distribution of the steps
- Independent Normal increments with variance equal to the elapsed time
- Continuous, nowhere differentiable, self-similar, a martingale
- We simulate it on a grid by summing $\sqrt{\Delta t} Z$ terms

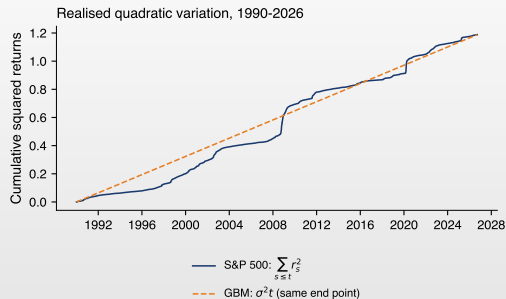
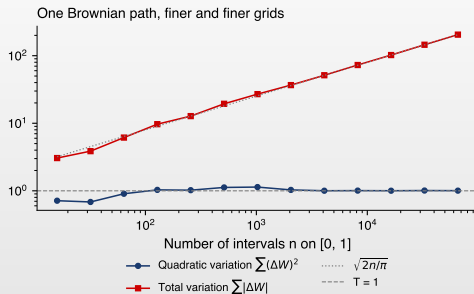
Total and Quadratic Variation

- Grid $0 = t_0 < t_1 < \dots < t_n = T$ with step $\Delta t = T/n$ and increments $\Delta W_k = W_{t_k} - W_{t_{k-1}}$
- **Total variation** $\sum_k |\Delta W_k|$: the length of the path
 - ▶ Its mean is $\sqrt{2nT/\pi} \rightarrow \infty$: a Brownian path has infinite length
- **Quadratic variation** $[W]_T = \lim \sum_k (\Delta W_k)^2 = T$
 - ▶ A random sum with a **deterministic** limit
 - ▶ For a differentiable function the quadratic variation is 0
- Shorthand used in Itô calculus: $(dW_t)^2 = dt$

Worked Example: Moments of the Quadratic Variation

- ▣ Each $(\Delta W_k)^2 = \Delta t Z_k^2$ with $Z_k \sim N(0, 1)$, so $\mathbb{E}[(\Delta W_k)^2] = \Delta t$ and $\text{Var}((\Delta W_k)^2) = 2\Delta t^2$
- ▣ Sum of n independent terms: $\mathbb{E}[Q_n] = n\Delta t = T$, $\text{Var}(Q_n) = 2n\Delta t^2 = 2T^2/n \rightarrow 0$
- ▣ One year of daily data, $T = 1$, $n = 252$: standard deviation of Q_n equals $\sqrt{2/252} = 0.089$
 - Squared returns estimate the variance of a year with about 9% relative error, if volatility is constant
- ▣ This is the idea behind realised variance from intraday data (Chapter 9)

Quadratic Variation: Simulation and Data



□ Left: one Brownian path on grids with 16 to 65,536 intervals; right: cumulative sum of squared daily S&P 500 log returns

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Reading the Quadratic Variation Chart

- ▣ Simulation: the quadratic variation settles at 1.006 ($T = 1$); the total variation keeps growing, 205 against $\sqrt{2n/\pi} = 204$
- ▣ S&P 500: total $\sum r_t^2 = 1.188$ over 36.7 years, an average volatility of 18.0%
 - ▶ Under geometric Brownian motion the curve would be a straight line
- ▣ It moves in steps: September 2008 – March 2009 is 1.6% of the days but 15.2% of all quadratic variation
 - ▶ February – April 2020: 0.5% of the days, 7.8% of the quadratic variation
- ▣ Volatility is itself random: the motivation for the Heston model at the end of the chapter

The Itô Integral

- Goal: give a meaning to $\int_0^T H_t dW_t$, the gain of holding H_t units of an asset whose price moves like W
 - ▶ Riemann–Stieltjes integration fails because the path has infinite total variation
- **Itô integral**: the limit of sums with the integrand at the **left** end point
 - ▶ $\int_0^T H_t dW_t = \lim_{n \rightarrow \infty} \sum_{k=1}^n H_{t_{k-1}} (W_{t_k} - W_{t_{k-1}})$ (limit in mean square)
- H must be **adapted**: H_t depends only on information up to t
 - ▶ Financial reading: you choose the position before you see the next price move

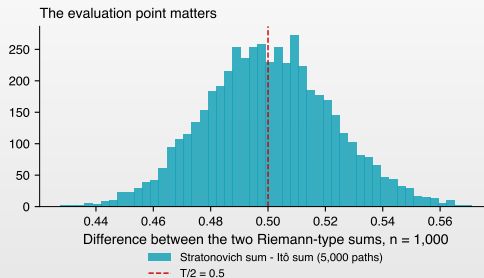
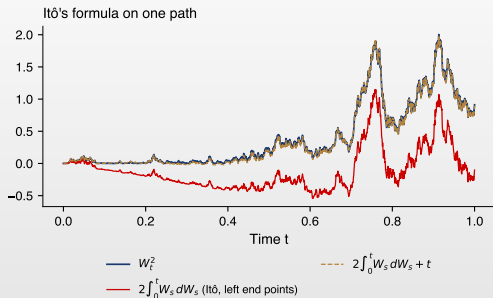
Properties of the Itô Integral

- ▣ Zero mean: $\mathbb{E} \left[\int_0^T H_t \, dW_t \right] = 0$: no strategy earns money from a martingale on average
- ▣ **Itô isometry**: $\mathbb{E} \left[\left(\int_0^T H_t \, dW_t \right)^2 \right] = \mathbb{E} \left[\int_0^T H_t^2 \, dt \right]$
- ▣ The process $t \mapsto \int_0^t H_s \, dW_s$ is a martingale
- ▣ The evaluation point matters: the mid-point rule gives the **Stratonovich** integral, a different number
 - ▶ Finance uses Itô, because a mid-point position would look into the future

Worked Example: $\int_0^T W_t dW_t$

- ▣ Identity for each term: $W_{t_{k-1}} \Delta W_k = \frac{1}{2} (W_{t_k}^2 - W_{t_{k-1}}^2) - \frac{1}{2} (\Delta W_k)^2$
- ▣ Sum over k : the first part telescopes to $\frac{1}{2} W_T^2$, the second tends to $\frac{1}{2} T$
 - ▶ $\int_0^T W_t dW_t = \frac{1}{2} W_T^2 - \frac{1}{2} T$
- ▣ Ordinary calculus would give $\frac{1}{2} W_T^2$; the extra $-\frac{1}{2} T$ is the quadratic variation
- ▣ Check: the mean is $\frac{1}{2}(T - T) = 0$, as it must be for an Itô integral

Itô's Formula on Simulated Paths



- Left: W_t^2 , the Itô integral $2\int_0^t W dW$ and their difference t on one path; right: Stratonovich minus Itô sums over 5,000 paths

Reading the Itô Chart

- Left: $2 \int_0^t W \, dW + t$ lies exactly on W_t^2 : the formula holds path by path, not only on average
- Right: Stratonovich minus Itô has mean 0.500 and standard deviation 0.022
 - ▶ The difference is half the quadratic variation, $T/2 = 0.5$
- Itô sums: mean 0.007, variance 0.520
 - ▶ Theory: mean 0 and, by the isometry, variance $\int_0^1 t \, dt = 1/2$

Itô Processes and Itô's Lemma

- **Itô process:** $dX_t = a_t dt + b_t dW_t$ (drift a_t , diffusion coefficient b_t , both adapted)
 - ▶ Shorthand for $X_t = X_0 + \int_0^t a_s ds + \int_0^t b_s dW_s$
- **Itô's lemma:** for $f(t, x)$ twice differentiable [Itô \(1944\)](#)
 - ▶ $df(t, X_t) = (f_t + a_t f_x + \frac{1}{2} b_t^2 f_{xx}) dt + b_t f_x dW_t$
- Rule of thumb: Taylor expansion to second order with $dt \cdot dt = 0$, $dt \cdot dW = 0$, $dW \cdot dW = dt$
 - ▶ The second-order term $\frac{1}{2} b_t^2 f_{xx} dt$ does not vanish: this is the whole novelty

Worked Example: the Logarithm of a Price

- ▣ Price dynamics $dS_t = \mu S_t dt + \sigma S_t dW_t$; take $f(x) = \ln x$, so $f_x = 1/x$, $f_{xx} = -1/x^2$
- ▣ Itô's lemma with $a_t = \mu S_t$, $b_t = \sigma S_t$
 - ▶ $d \ln S_t = \left(\mu S_t \cdot \frac{1}{S_t} - \frac{1}{2} \sigma^2 S_t^2 \cdot \frac{1}{S_t^2} \right) dt + \sigma S_t \cdot \frac{1}{S_t} dW_t = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t$
- ▣ The log price grows at $\mu - \sigma^2/2$, not at μ : volatility lowers the growth of the median investor
- ▣ With $\sigma = 18\%$: $\sigma^2/2 = 1.6$ percentage points per year; with $\sigma = 67\%$ (Bitcoin): 22 points

Recap: Itô Calculus

- Brownian paths have infinite length but finite quadratic variation $[W]_t = t$
- The Itô integral uses left end points: it is a martingale with zero mean
- Itô's lemma = Taylor to second order with $(dW)^2 = dt$
- Real prices have a quadratic variation that grows in bursts, not linearly

Stochastic Differential Equations

- A **stochastic differential equation** (SDE): $dX_t = a(t, X_t) dt + b(t, X_t) dW_t$, X_0 given
 - ▶ Drift a and diffusion b now depend on the current state
- Existence and uniqueness of a solution: a and b Lipschitz in x with at most linear growth [Kloeden & Platen \(1992\)](#)
- Few SDEs have closed-form solutions; the four models of this chapter are among them
 - ▶ Geometric Brownian motion, Ornstein–Uhlenbeck and Merton: exact; Heston: only the characteristic function
- Otherwise we simulate on a grid; the question is how large the discretisation error is

Euler–Maruyama and Milstein Schemes

- **Euler–Maruyama:** freeze a and b over each step [Higham \(2001\)](#)

- ▶ $X_{k+1} = X_k + a(X_k) \Delta t + b(X_k) \Delta W_k, \Delta W_k = \sqrt{\Delta t} Z_k$

- **Milstein:** add the next Itô–Taylor term [Milstein \(1975\)](#)

- ▶ $X_{k+1} = X_k + a(X_k) \Delta t + b(X_k) \Delta W_k + \frac{1}{2} b(X_k) b'(X_k) [(\Delta W_k)^2 - \Delta t]$

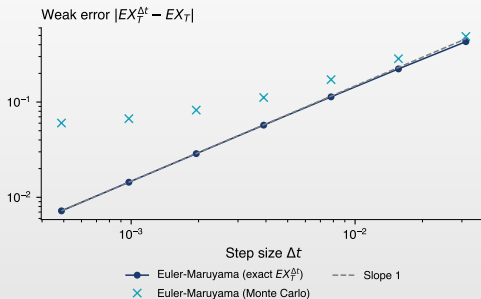
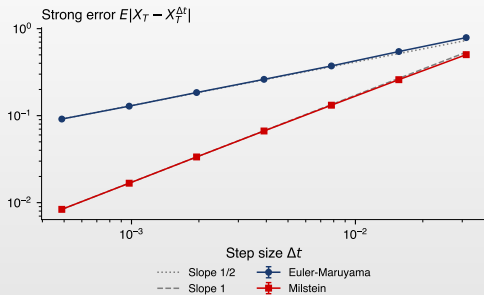
- The correction uses $(\Delta W_k)^2 - \Delta t$, which has mean zero: it only matters path by path

- Test equation [Higham \(2001\)](#): $dX = \lambda X dt + \mu X dW$, $\lambda = 2$, $\mu = 1$, $X_0 = 1$, $T = 1$, with the exact solution as benchmark

Strong and Weak Convergence

- **Strong error:** $\mathbb{E}|X_T - X_T^{\Delta t}| \leq C \Delta t^\gamma$, how close each simulated path is to the true one
 - ▶ Euler–Maruyama $\gamma = 1/2$; Milstein $\gamma = 1$
 - ▶ Needed for path-dependent quantities: hedging errors, barrier crossings
- **Weak error:** $|\mathbb{E}[g(X_T^{\Delta t})] - \mathbb{E}[g(X_T)]| \leq C \Delta t^\beta$, how close the distribution is
 - ▶ Both schemes: $\beta = 1$; enough for prices of European options, VaR, expected values
- On a log-log plot, the order is the slope of error against step size

Convergence of the Schemes



□ 20,000 Brownian paths on a fine grid of 2^{11} steps, coarsened to steps from 2^{-11} to 2^{-5} ; bars are 95% CIs

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Reading the Convergence Chart

- ▣ Estimated strong orders (95% CI): Euler–Maruyama 0.518 [0.509, 0.528], Milstein 0.986 [0.977, 0.995]
 - ▶ Close to the theoretical $1/2$ and 1 ; halving Δt cuts the Milstein error in half, the Euler error only by 29%
- ▣ Weak order of Euler–Maruyama from the exact mean $(1 + \lambda \Delta t)^{T/\Delta t}$: 0.985
 - ▶ The Monte Carlo estimates flatten at small steps: their standard error, about 0.068, exceeds the bias
- ▣ In practice the sampling error of Monte Carlo, not the step size, is often the binding constraint [Glasserman \(2003\)](#)

Recap: Simulation

- An SDE specifies the local drift and diffusion of a process
- Euler–Maruyama is simple and strong order $1/2$; Milstein adds one term and reaches order 1
- For expected values the weak order 1 is what matters, together with the Monte Carlo sample size
- When the exact transition is known (next sections), simulate exactly: no discretisation error

Geometric Brownian Motion (GBM)

- SDE: $dS_t = \mu S_t dt + \sigma S_t dW_t$, i.e. the **percentage** change has constant drift and volatility
 - ▶ μ : expected rate of return; σ : volatility, both per year
- Solution (from the log example): $S_t = S_0 \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right)$
 - ▶ Prices stay positive; exact simulation needs only Normal draws
- The price model behind [Black & Scholes \(1973\)](#); proposed by [Osborne \(1959\)](#) and [Samuelson \(1965\)](#)
- Textbook treatment: [Franke, Härdle & Hafner \(2019\)](#), chapters on the Wiener process and stochastic integration

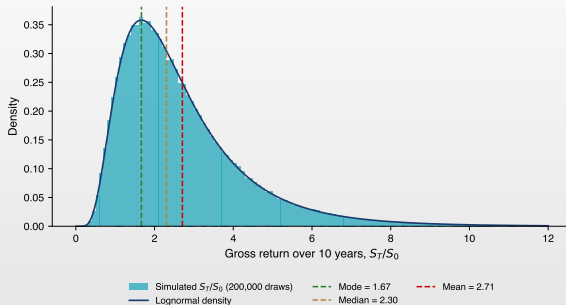
Log Returns under GBM

- Log returns over Δt : $r_t = \ln(S_t/S_{t-\Delta t}) \sim N((\mu - \frac{1}{2}\sigma^2)\Delta t, \sigma^2\Delta t)$
 - ▶ Independent and identically distributed (i.i.d.) at every frequency
- Testable implications
 - ▶ Normal log returns: skewness 0, excess kurtosis 0
 - ▶ No autocorrelation in r_t , $|r_t|$ or r_t^2
 - ▶ Variance proportional to the horizon
- S_T/S_0 is **lognormal**: $\ln(S_T/S_0) \sim N(mT, \sigma^2 T)$ with the log drift $m = \mu - \sigma^2/2$

Mean, Median and Mode of a Lognormal Price

- Mean $\mathbb{E}[S_T] = S_0 e^{\mu T}$; median $S_0 e^{mT}$; mode $S_0 e^{(m-\sigma^2)T}$: mode < median < mean
- Worked example, S&P 500 parameters: $\hat{\mu} = 10.0\%$, $\hat{\sigma} = 18.0\%$, $T = 10$ years
 - ▶ Mean gross return 2.71, median 2.30, mode 1.67
 - ▶ $P(S_T < S_0) = \Phi(-m\sqrt{T}/\sigma) = 7.1\%$
 - ▶ $P(S_T < \mathbb{E}[S_T]) = \Phi(\sigma\sqrt{T}/2) = 61.2\%$: most investors end **below** the average
- The average is pulled up by a few very large outcomes

The Lognormal Distribution after Ten Years

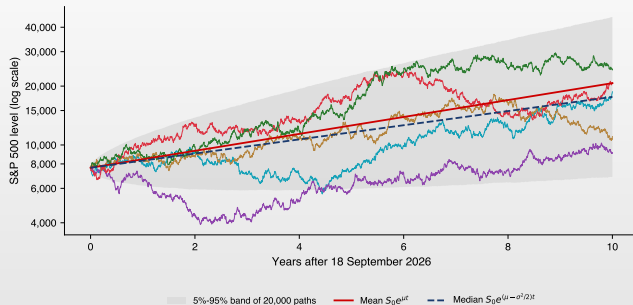


- Gross return S_T/S_0 over 10 years under GBM with the S&P 500 parameters: 200,000 exact draws and the lognormal density

Reading the Lognormal Chart

- The distribution is skewed to the right: long upper tail, bounded below by zero
- The most likely outcome (1.67) is well below the mean (2.71)
 - ▶ Advertised “average returns” overstate what a typical investor experiences
- The gap grows with $\sigma^2 T$: for Bitcoin, with σ near 67%, the median is far below the mean
- For long horizons, report the median and quantiles, not only the mean

Ten Years of Simulated S&P 500 Paths



- GBM with $\hat{\mu}$, $\hat{\sigma}$ from 1990–2026, starting at the close of 18 September 2026 ($S_0 = 7\,650$); 20,000 exact paths, five shown

Reading the Fan Chart

- After ten years: mean 20 707, median 17 612
 - ▶ 5%–95% range: 6 879 – 45 049; the upper tail is three times further from the median than the lower one
- Probability of ending below today's level: 7.1%
- Individual paths show long drawdowns even when the drift is positive
- This fan is too narrow in crises and too smooth: GBM has no fat tails and no volatility clustering
 - ▶ Next slides: estimating the parameters and testing GBM against the data

Estimating GBM by Maximum Likelihood

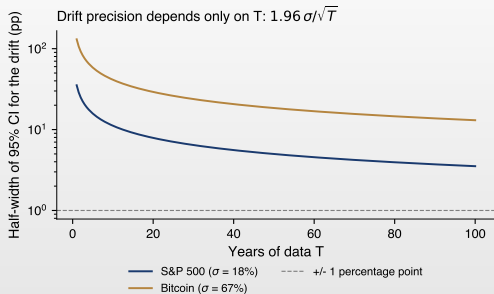
- Maximum likelihood estimation (MLE) from n log returns at step Δt , over $T = n\Delta t$ years
 - ▶ $\hat{\sigma}^2 = \frac{1}{n\Delta t} \sum_t (r_t - \bar{r})^2$, $\hat{m} = \bar{r}/\Delta t$, $\hat{\mu} = \hat{m} + \hat{\sigma}^2/2$
- Standard errors: $\text{s.e.}(\hat{\sigma}) = \sigma/\sqrt{2n}$ and $\text{s.e.}(\hat{m}) = \sigma/\sqrt{T}$
 - ▶ Volatility precision improves with more observations (sample more often)
 - ▶ Drift precision depends only on the **calendar length** T : daily or minute data do not help
- Reason: $\sum_t r_t = \ln(S_T/S_0)$ uses only the first and the last price

Worked Example: GBM for Three Markets

	Years	$\hat{\sigma}$ (%)	s.e. ($\hat{\sigma}$)	$\hat{m}$ (%)	s.e. ($\hat{m}$)
S&P 500	36.7	18.0	0.13	8.3	3.0
BET	26.5	22.2	0.19	16.1	4.3
Bitcoin	12.0	66.6	0.71	43.1	19.2

- Volatility is pinned down to about 0.1–0.7 percentage points; the log drift only to 3–19 points
- S&P 500: $\hat{\mu} = \hat{m} + \hat{\sigma}^2/2 = 8.3\% + 1.62\% = 10.0\%$ per year
- A 95% CI of ± 1 point for the S&P 500 drift would need $(1.96 \hat{\sigma}/0.01)^2 = 1\,244$ years of data

How Precisely Can We Estimate the Drift?



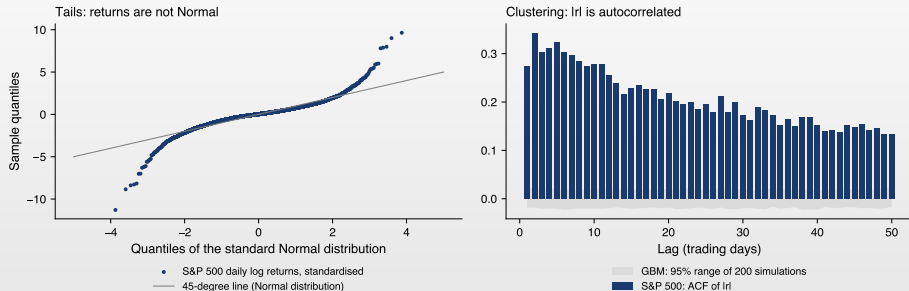
□ Left: half-width of the 95% CI for the drift against years of data; right: S&P 500 estimates on rolling 10-year windows

 MFM_ch11_drift_precision

Reading the Drift Chart

- Ten-year log drift of the S&P 500 ranges from -6.1% (window ending 2009-03) to 15.8% (2000-10)
 - ▶ Every window's 95% CI is about ± 11 points wide: the variation is mostly noise
- Ten-year volatility stays between 14.1% and 22.3% : estimated precisely, but not constant
- Bitcoin would need 17 028 years for ± 1 point: its expected return is essentially unknowable from its price history
- Option pricing in Chapter 12 avoids μ altogether: only σ enters the Black–Scholes price

Does GBM Fit the S&P 500?



- Left: quantile–quantile (QQ) plot of standardised daily log returns against the Normal distribution; right: autocorrelation function (ACF) of $|r_t|$ against 200 GBM samples of the same length

 MFM_ch11_gbm_vs_data

Reading the GBM Test

- Tails: excess kurtosis 10.9 instead of 0; the worst day is -11.3 standard deviations
 - ▶ 62 days beyond ± 4 standard deviations, against 0.59 expected under GBM
 - ▶ Jarque–Bera (JB) statistic 45 718: Normality is rejected at any level
- Clustering: ACF of $|r_t|$ is 0.27 at lag 1 and still 0.13 at lag 50
 - ▶ GBM samples never exceed 0.024 in absolute value
- GBM fails on both stylised facts of Chapter 1 [Cont \(2001\)](#); the rest of the chapter adds mean reversion, jumps and random volatility

Recap: Geometric Brownian Motion

- GBM: constant percentage drift and volatility; lognormal prices; i.i.d. Normal log returns
- The median grows at $\mu - \sigma^2/2$; the mean at μ ; most paths end below the mean
- Volatility is estimable from data; the drift is not, in any realistic sample
- The data reject GBM: fat tails and volatility clustering

The Ornstein–Uhlenbeck (OU) Process

- SDE: $dx_t = \kappa(\theta - x_t) dt + \sigma dW_t$, $\kappa > 0$ Uhlenbeck & Ornstein (1930)
 - ▶ θ : long-run mean; κ : speed of mean reversion; σ : volatility
 - ▶ Above θ the drift is negative, below θ positive: a spring pulling x back
- Used for interest rates, volatility, spreads, commodity prices, exchange-rate deviations
 - ▶ Vasicek (1977): the OU process as a model of the short-term interest rate

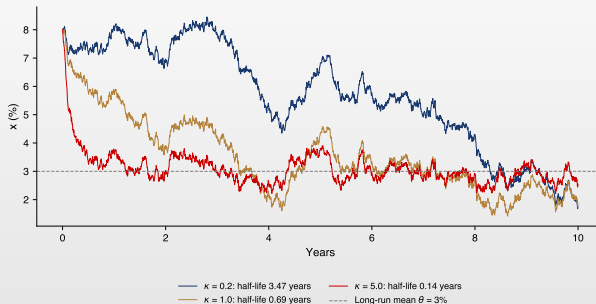
Solution and Stationary Distribution

- By Itô's lemma on $e^{\kappa t} x_t$: $x_t = \theta + (x_0 - \theta)e^{-\kappa t} + \sigma \int_0^t e^{-\kappa(t-s)} dW_s$
- Conditional distribution: Normal with
 - ▶ $\mathbb{E}[x_t | x_0] = \theta + (x_0 - \theta)e^{-\kappa t}$, $\text{Var}(x_t | x_0) = \frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa t})$
- Stationary distribution ($t \rightarrow \infty$): $N(\theta, \sigma^2/(2\kappa))$
- **Half-life** $\ln 2/\kappa$: the time for an expected deviation from θ to halve
 - ▶ $\kappa = 0.5$ per year: half-life 1.39 years

Exact Discretisation: an AR(1)

- Sampled at step Δt the OU process is exactly an autoregressive model of order 1 (AR(1)):
 - ▶ $x_{t+\Delta t} = a + b x_t + \varepsilon_t$, $b = e^{-\kappa \Delta t}$, $a = \theta(1 - b)$, $\text{Var}(\varepsilon_t) = \frac{\sigma^2}{2\kappa}(1 - b^2)$
- Maximum likelihood = ordinary least squares (OLS) of $x_{t+\Delta t}$ on x_t , then invert the mapping
 - ▶ $\hat{\kappa} = -\ln \hat{b} / \Delta t$, $\hat{\theta} = \hat{a} / (1 - \hat{b})$, $\hat{\sigma} = \hat{s}_\varepsilon \sqrt{2\hat{\kappa} / (1 - \hat{b}^2)}$
- Standard errors for $\hat{\kappa}$ and $\hat{\theta}$ by the delta method from the OLS covariance of $(\hat{a}, \hat{b})$
- With daily data b is very close to 1: κ is hard to estimate, as with unit-root tests

Three Speeds of Mean Reversion



- OU paths driven by the same shocks, starting at 8%, with $\theta = 3\%$, $\sigma = 1\%$ and $\kappa = 0.2, 1, 5$; exact simulation, daily steps

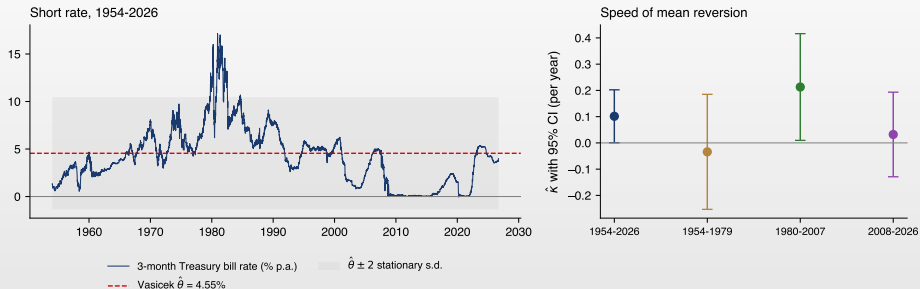
Reading the OU Chart

- Half-lives: 3.47, 0.69 and 0.14 years
 - ▶ With $\kappa = 5$ the gap from 8% to 3% closes within a few months
- Stationary standard deviations $\sigma/\sqrt{2\kappa}$: 1.58%, 0.71%, 0.32%
 - ▶ Fast reversion keeps the process in a narrow band; slow reversion looks like a random walk for years
- Unlike Brownian motion, the variance stops growing: long-horizon forecasts converge to θ

Short-Rate Models: Vasicek and CIR

- **Vasicek**: the short rate r_t is an OU process [Vasicek \(1977\)](#)
 - ▶ Normal distribution: negative rates have positive probability $\Phi(-\theta\sqrt{2\kappa}/\sigma)$ in the long run
 - ▶ Bond prices and yields in closed form
- **Cox–Ingersoll–Ross (CIR)**: $dr_t = \kappa(\theta - r_t) dt + \sigma\sqrt{r_t} dW_t$ [Cox, Ingersoll & Ross \(1985\)](#)
 - ▶ Volatility shrinks near zero; the rate stays positive if $2\kappa\theta \geq \sigma^2$ (Feller condition) [Feller \(1951\)](#)
- Which is closer to the data is an empirical question [Chan, Karolyi, Longstaff & Sanders \(1992\)](#); exact likelihoods for general diffusions: [Aït-Sahalia \(2002\)](#)

Vasicek on the 3-Month Treasury Bill Rate



- Left: daily 3-month US Treasury bill rate (DTB3), 1954–2026, with the Vasicek long-run mean; right: $\hat{\kappa}$ with 95% CI by period

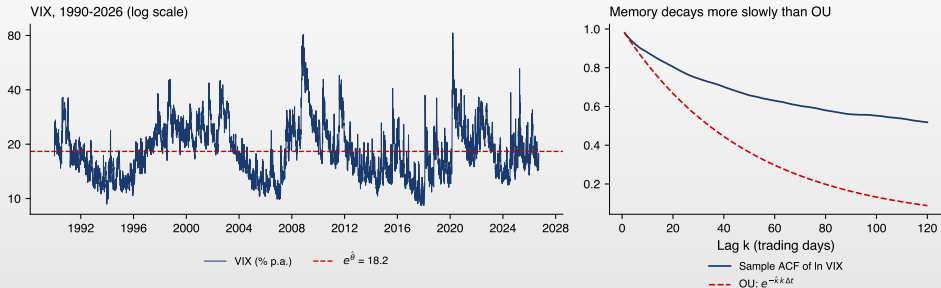
Reading the Vasicek Chart

- Full sample: $\hat{\theta} = 4.55\%$, $\hat{\kappa} = 0.101$ (s.e. 0.051), half-life 6.8 years, $\hat{\sigma} = 1.33\%$
 - ▶ Daily AR(1) coefficient $\hat{b} = 0.99960$: almost a unit root
- Subperiods disagree: $\hat{\kappa} = -0.034$ (1954–1979, explosive), 0.213 (1980–2007), 0.032 (2008–2026)
 - ▶ Two of the three 95% CIs include zero: mean reversion is not detectable within 20–25 years
- The rate ranged from -0.05% (2020-03-26) to 17.1% (1980-12-11); 4.3% of days were below 0.05%
 - ▶ The fitted Vasicek model gives a 6.1% long-run probability of a negative rate

The Mean-Reversion Speed Is Biased Upwards

- In finite samples $\hat{\kappa}$ overstates κ ; the bias is roughly $4/T$ with T in years [Tang & Chen \(2009\)](#)
 - ▶ It depends on the calendar span, not on the sampling frequency, like the drift of GBM
- Simulation with the full-sample Vasicek estimates as the truth (72 years, 500 samples)
 - ▶ Mean $\hat{\kappa} = 0.164$ against the true 0.101: bias 0.063, close to $4/T = 0.055$
 - ▶ 78% of the estimates exceed the truth
- Consequence: estimated half-lives are too short, and mean reversion looks faster than it is

Is Volatility Mean-Reverting? The VIX



- Left: VIX 1990–2026 on a log scale with the OU long-run level of $\ln VIX$; right: sample ACF of $\ln VIX$ against the OU-implied $e^{-\hat{\kappa} k \Delta t}$

Reading the VIX Chart

- OU fit to $\ln VIX$: $\hat{\kappa} = 5.10$ (s.e. 0.53), half-life 34 trading days, long-run level $e^{\hat{\theta}} = 18.2$
 - ▶ Volatility spikes (VIX 82.7 on 2020-03-16) fade within weeks, not years
- But the ACF at lag 60 is 0.63, against 0.30 implied by OU
 - ▶ Fast decay first, then a slow component: one OU factor is too simple
 - ▶ Two-factor or long-memory models (rough volatility [Gatheral, Jaisson & Rosenbaum \(2018\)](#)) fit better
- Mean-reverting volatility is the building block of the Heston model

Recap: Mean Reversion

- OU: a linear pull towards θ with speed κ ; Normal, stationary, exactly an AR(1) when sampled
- Half-life $\ln 2 / \kappa$ summarises the persistence
- Interest rates: κ small, unstable across periods and biased upwards in finite samples
- Volatility (VIX): clear mean reversion within weeks, plus a slow component

Why Jumps?



Amsterdam Stock Exchange, 21 October 1987, after the crash of 19 October

Photo: Bart Molendijk / Anefo (1987); CC0; Wikimedia Commons

- On 19 October 1987 the Dow Jones fell 22.6% in one day
 - ▶ Under GBM with 18% volatility this is a move of about 20 standard deviations
- Brownian paths are continuous: large moves need many small steps
- Real prices gap on news: earnings, central-bank decisions, defaults, exchange hacks
- A jump cannot be hedged by continuous trading: this breaks the Black–Scholes argument [Merton \(1976\)](#)

Poisson and Compound Poisson Processes

- **Poisson process** N_t with intensity λ : counts events; increments independent, $N_{t+h} - N_t \sim \text{Poisson}(\lambda h)$
 - ▶ Expected number of jumps per year: λ ; waiting times exponential with mean $1/\lambda$
 - ▶ Over one day: $P(\text{no jump}) = e^{-\lambda \Delta t}$, $P(\text{one jump}) \approx \lambda \Delta t$
- **Compound Poisson**: $J_t = \sum_{i=1}^{N_t} Y_i$ with jump sizes Y_i i.i.d., independent of N
 - ▶ Merton: $Y_i \sim N(\mu_J, \sigma_J^2)$ are jumps in the **log** price

Merton's Jump-Diffusion

- ▣ Log price = Brownian motion with drift + compound Poisson jumps [Merton \(1976\)](#)
 - ▶ $d \ln S_t = m dt + \sigma dW_t + dJ_t$, $J_t = \sum_{i \leq N_t} Y_i$
- ▣ Return over Δt given k jumps: $N(m\Delta t + k\mu_J, \sigma^2\Delta t + k\sigma_J^2)$
 - ▶ Unconditional density: a **Poisson mixture** of Normal densities
 - ▶ $f(r) = \sum_{k \geq 0} e^{-\lambda\Delta t} \frac{(\lambda\Delta t)^k}{k!} \varphi(r; m\Delta t + k\mu_J, \sigma^2\Delta t + k\sigma_J^2)$
- ▣ Five parameters ($m, \sigma, \lambda, \mu_J, \sigma_J$); returns still i.i.d. over time
- ▣ Exact simulation: draw $N \sim \text{Poisson}(\lambda\Delta t)$, then a Normal with the conditional moments

Moments of the Merton Return

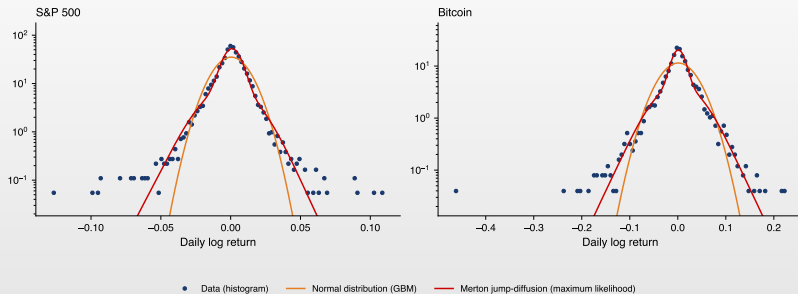
- Cumulants of the return over Δt (the k -th cumulant of a compound Poisson sum is $\lambda \Delta t \mathbb{E}[Y^k]$):
 - ▶ $\kappa_2 = \sigma^2 \Delta t + \lambda \Delta t (\mu_J^2 + \sigma_J^2)$, $\kappa_3 = \lambda \Delta t (\mu_J^3 + 3\mu_J \sigma_J^2)$
 - ▶ $\kappa_4 = \lambda \Delta t (\mu_J^4 + 6\mu_J^2 \sigma_J^2 + 3\sigma_J^4)$; skewness $\kappa_3 / \kappa_2^{3/2}$, excess kurtosis κ_4 / κ_2^2
- Negative μ_J gives negative skewness; any jump gives positive excess kurtosis
- Excess kurtosis falls like $1/\Delta t$ as the horizon grows: jumps matter most at short horizons
 - ▶ Example: $\sigma = 15\%$, $\lambda = 5$, $\mu_J = -3\%$, $\sigma_J = 4\%$: daily excess kurtosis 17.6, skewness -2.07

Maximum Likelihood Estimates

	$\hat{\sigma}$ (%)	$\hat{\lambda}$ (per year)	$\hat{\mu}_J$ (%)	$\hat{\sigma}_J$ (%)	Jump share of variance	LR
S&P 500	9.5	101 (10)	-0.16	1.45	71%	2 397
BET	11.3	79 (7)	-0.09	2.07	73%	2 326
Bitcoin	24.6	239 (21)	0.00	3.85	85%	1 357

- Log-likelihood maximised numerically over the Poisson mixture (up to 12 jumps per day), standard errors from the Hessian in brackets
- LR: likelihood-ratio statistic against GBM ($\lambda = 0$)
- The fit prefers **many small** jumps (about 80–240 per year), not a few crashes

Merton Fit to the Return Distribution



- Density of daily log returns on a log scale: histogram, the Normal distribution implied by GBM and the fitted Merton mixture

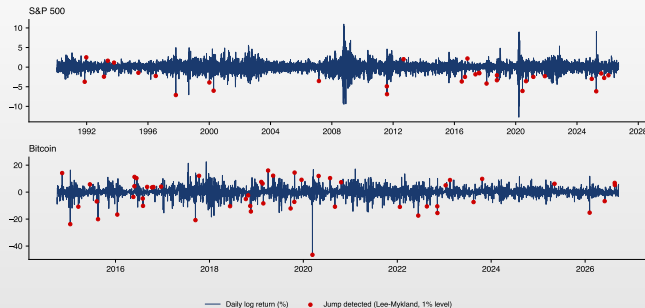
Reading the Merton Fit

- The Merton density follows the peak and the shoulders far better than the Normal distribution
 - ▶ LR statistics in the thousands: GBM is overwhelmingly rejected
- S&P 500: implied excess kurtosis 3.7 against 10.9 in the data; skewness -0.31 against -0.36
 - ▶ The most extreme days lie beyond both fitted curves
- About 101 “jumps” per year means the model uses jumps to mimic **changing volatility**
 - ▶ An i.i.d. model cannot tell a jump from a volatile period: we need a direct test for jumps

Detecting Jumps: the Lee–Mykland Test

- Idea: a jump is a return that is huge **relative to the local volatility** Lee & Mykland (2008)
 - ▶ $L_t = r_t / \hat{\sigma}_t$, $\hat{\sigma}_t^2 = \frac{1}{K-2} \sum_{j=t-K+2}^{t-1} |r_j| |r_{j-1}|$ (bipower variation, robust to jumps)
- Under no jumps, the maximum of $|L_t|$ over n days has a Gumbel limit
 - ▶ Reject if $(|L_t| - C_n) / S_n > -\ln(-\ln(1 - \alpha))$, with C_n, S_n depending on n ; at $\alpha = 1\%$ the threshold is 4.60
- Daily data: window $K = 16$ days, as recommended by the authors
- Controls the probability of a false jump over the **whole** sample, not day by day

Jumps Detected in the S&P 500 and Bitcoin



□ Daily log returns (%) with the days flagged by the Lee–Mykland test at the 1% level

Reading the Jump Chart

- ▣ S&P 500: 32 jumps in 36.7 years, 0.87 per year; 84% of them are falls
 - ▶ Not one of the 10 largest absolute returns is flagged: they came in turbulent periods (2008, 2020)
 - ▶ Instead, moves as small as 1.2% in a very calm market are jumps (1993-10-14)
- ▣ Bitcoin: 52 jumps, 4.35 per year, 5 of its 10 largest moves flagged
 - ▶ Up and down jumps equally often (52% negative)
- ▣ True jumps are rare (about 1 to 4 per year), far fewer than the Merton MLE suggests: the rest is stochastic volatility

Recap: Jumps

- Merton adds compound Poisson jumps to GBM; returns become a Poisson mixture of Normal distributions
- Jumps create skewness and excess kurtosis, strongest at short horizons
- Fitted by MLE on i.i.d. returns, the model mistakes volatile periods for frequent jumps
- Nonparametric tests relative to local volatility find only a few jumps per year

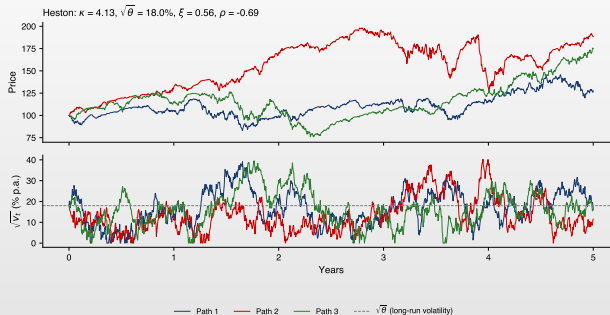
The Heston Model

- Two SDEs, for the log price and for the instantaneous variance v_t [Heston \(1993\)](#)
 - ▶ $d \ln S_t = (\mu - \frac{1}{2} v_t) dt + \sqrt{v_t} dW_t^{(1)}$
 - ▶ $dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^{(2)}, \text{Corr}(dW^{(1)}, dW^{(2)}) = \rho$
- Variance follows a CIR process: mean-reverting to θ with speed κ ; ξ = volatility of volatility
 - ▶ Feller condition $2\kappa\theta \geq \xi^2$: variance never touches zero [Feller \(1951\)](#)
- $\rho < 0$: **leverage effect**, falling prices come with rising volatility
 - ▶ Simulation: Euler scheme with full truncation $v^+ = \max(v, 0)$ [Lord, Koekkoek & van Dijk \(2010\)](#); exact alternatives [Andersen \(2008\)](#)

Heston Parameters from the VIX

- Proxy for the variance: $v_t \approx (\text{VIX}_t/100)^2$; discretised CIR regression
 - ▶ $\frac{v_{t+\Delta t} - v_t}{\sqrt{v_t}} = \kappa\theta \frac{\Delta t}{\sqrt{v_t}} - \kappa\Delta t\sqrt{v_t} + \xi\sqrt{\Delta t}\varepsilon_t$; ρ = correlation of return and variance shocks
- S&P 500 and VIX, 1990–2026: $\hat{\kappa} = 4.13$ (half-life 42 days), $\hat{\xi} = 0.56$, $\hat{\rho} = -0.69$
 - ▶ $\sqrt{\hat{\theta}} = 20.9\%$ from the VIX, but the realised volatility is 18.0%
 - ▶ Feller ratio $2\hat{\kappa}\hat{\theta}/\hat{\xi}^2 = 1.16$
- The VIX is a **risk-neutral** expectation: it includes a variance risk premium
 - ▶ For simulating real-world paths we set θ to the realised variance; for option prices we keep the VIX value

Simulated Heston Paths

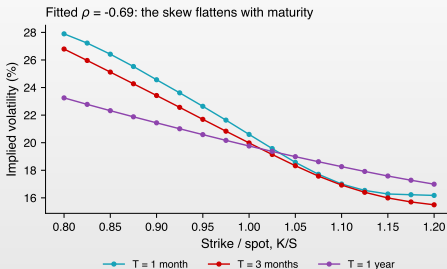
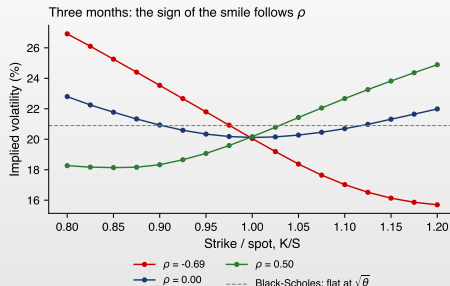


- Three five-year paths with the VIX-based parameters and θ equal to the realised S&P 500 variance; top: price, bottom: $\sqrt{V_t}$

Reading the Heston Paths

- Volatility wanders between near zero and about 40%, with calm and turbulent episodes
 - ▶ Volatility clustering now comes from the model, not from an external assumption
- Price falls coincide with volatility spikes: the simulated correlation of return and variance changes is -0.69
 - ▶ This is the leverage effect built in through ρ
- With the Feller ratio only slightly above 1, the variance comes close to zero from time to time
 - ▶ The discretisation must handle $v_t < 0$; full truncation does this

A Preview of Chapter 12: the Volatility Smile



- Black–Scholes implied volatilities of Heston call prices (Monte Carlo, 100,000 antithetic paths); left: effect of ρ at 3 months; right: maturities with the fitted ρ

Reading the Smile Chart

- ▣ **Implied volatility**: the σ that makes the Black–Scholes price equal to a given option price
 - ▶ Under GBM it is the same for every strike: a flat line
- ▣ With $\hat{\rho} = -0.69$, three months: 26.9% at strike 80% of spot, 20.0% at the money, 15.7% at 120%
 - ▶ A downward **skew**, as in equity-index options; with $\rho = 0$ a symmetric **smile** (22.8%, 20.1%, 22.0%)
- ▣ The skew flattens with maturity: from 27.9% – 16.2% at one month to 23.3% – 17.0% at one year
- ▣ Chapter 12 fits such surfaces to market option prices

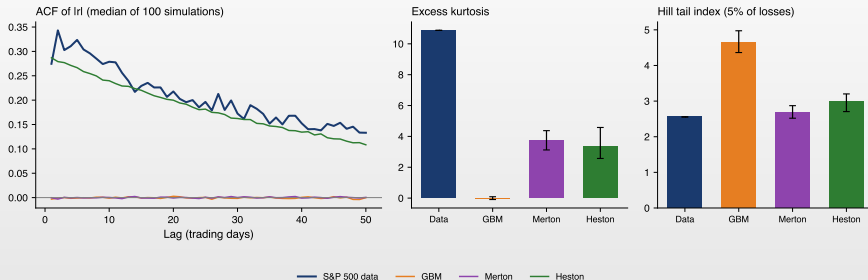
From GARCH to Continuous Time

- Nelson (1990): as the time step shrinks, GARCH(1,1) converges to a continuous-time stochastic-volatility diffusion
 - ▶ $dv_t = \kappa(\theta - v_t)dt + \xi v_t dW_t^{(2)}$: like Heston, but with v_t instead of $\sqrt{v_t}$
- The GARCH persistence $\alpha + \beta$ of Chapter 5 maps to $\kappa \approx (1 - \alpha - \beta)/\Delta t$
 - ▶ $\alpha + \beta = 0.99$ with daily data gives $\kappa \approx 2.5$ per year, a half-life of about 70 days
- Discrete-time GARCH for forecasting and risk; continuous-time stochastic volatility for pricing and hedging
- Further reading: Čížek, Härdle & Weron (2011) (Heston model chapter), Härdle, Chen & Overbeck (2017)

Recap: Stochastic Volatility

- ▣ Heston: variance is a mean-reverting CIR process correlated with returns
- ▣ Parameters can be read from the VIX: fast mean reversion, high volatility of volatility, strong leverage
- ▣ Generates volatility clustering and, through ρ , the implied-volatility skew
- ▣ GARCH is its discrete-time cousin

Which Model Reproduces the S&P 500?



- 100 simulated samples of 9,240 days from each fitted model: median ACF of $|r_t|$, excess kurtosis and Hill tail index with 90% ranges

Reading the Model Comparison

	Data	GBM	Merton	Heston
Excess kurtosis	10.87	0.00	3.72	3.34
Skewness	-0.36	0.00	-0.30	0.01
Hill tail index	2.56	4.64	2.68	2.98
ACF of $ r $, lag 1	0.27	0.00	0.00	0.29
ACF of $ r $, sum of lags 1–20	5.32	0.00	0.00	4.77

- Medians of 100 simulations; Hill index from the 5% largest losses (lower = heavier tail)
- Merton gets the tail index and skewness but no clustering; Heston gets the clustering but no skewness
- No model reaches the kurtosis of 10.9: it needs jumps **and** stochastic volatility, as in the SVJ model of [Bates \(1996\)](#)

Which Model When?

Model	Captures	Typical use
Geometric Brownian motion	Positive prices, $\sqrt{t}$ scaling	Black–Scholes benchmark, long-horizon planning
Ornstein–Uhlenbeck, CIR	Mean reversion	Interest rates, spreads, volatility indices
Merton jump-diffusion	Gaps, short-horizon fat tails	Short-dated options, credit events, crypto
Heston stochastic volatility	Clustering, leverage, skew	Equity-index options, volatility trading
Bates (Heston + jumps)	All of the above	Option surfaces across maturities

Beyond This Chapter

- ▣ Heavier tails: Lévy processes and α -stable distributions [Pele \(2014\)](#)
 - ▶ Crypto-assets form a distinct class by the shape of their return distributions [Pele et al. \(2023\)](#)
- ▣ Rough volatility: volatility paths rougher than Brownian motion [Gatheral, Jaisson & Rosenbaum \(2018\)](#)
 - ▶ High-frequency information about Bitcoin volatility and tail risk [Pele & Mazurencu-Marinescu-Pele \(2019\)](#)
- ▣ Exercises with solutions on all models of this chapter: [Borak, Härdle & López-Cabrera \(2013\)](#)
- ▣ Next, Chapter 12: pricing options by replication and reading the volatility surface

Summary

- Brownian motion is the universal limit of random walks; its quadratic variation t drives Itô calculus
- Itô's lemma turns SDEs into explicit solutions: GBM, Ornstein–Uhlenbeck, Merton
- Simulation: Euler–Maruyama and Milstein, or exact transitions when they exist
- Estimation: volatility is easy, drift and mean-reversion speed are hard and biased
- Data: jumps explain the tails, stochastic volatility the clustering; realistic models need both

Further Study on Quantinar

Advanced courses on quantinar.com

-  Applied Computational Finance
-  XFG Advanced Methods in Quantitative Finance

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_11

References (1/3)

- ▣ Aït-Sahalia, Y. (2002). Maximum likelihood estimation of discretely sampled diffusions: a closed-form approximation approach. *Econometrica*, 70(1), 223–262.
- ▣ Andersen, L. (2008). Simple and efficient simulation of the Heston stochastic volatility model. *Journal of Computational Finance*, 11(3), 1–42.
- ▣ Bachelier, L. (1900). Théorie de la spéculation. *Annales scientifiques de l'École Normale Supérieure*, 17, 21–86.
- ▣ Bates, D.S. (1996). Jumps and stochastic volatility: exchange rate processes implicit in Deutsche Mark options. *Review of Financial Studies*, 9(1), 69–107.
- ▣ Billingsley, P. (1999). *Convergence of Probability Measures*, 2nd ed. Wiley.
- ▣ Black, F., Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3), 637–654.
- ▣ Borak, S., Härdle, W.K., López-Cabrera, B. (2013). *Statistics of Financial Markets: Exercises and Solutions*, 2nd ed. Springer.
- ▣ Chan, K.C., Karolyi, G.A., Longstaff, F.A., Sanders, A.B. (1992). An empirical comparison of alternative models of the short-term interest rate. *Journal of Finance*, 47(3), 1209–1227.
- ▣ Čížek, P., Härdle, W.K., Weron, R. (Eds.) (2011). *Statistical Tools for Finance and Insurance*, 2nd ed. Springer.
- ▣ Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- ▣ Cox, J.C., Ingersoll, J.E., Ross, S.A. (1985). A theory of the term structure of interest rates. *Econometrica*, 53(2), 385–407.
- ▣ Einstein, A. (1905). Über die von der molekularkinetischen Theorie der Wärme geforderte Bewegung von in ruhenden Flüssigkeiten suspendierten Teilchen. *Annalen der Physik*, 322(8), 549–560.

References (2/3)

- ▣ Feller, W. (1951). [Two singular diffusion problems](#). *Annals of Mathematics*, 54(1), 173–182.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). [Statistics of Financial Markets: An Introduction](#), 5th ed. Springer.
- ▣ Gatheral, J., Jaisson, T., Rosenbaum, M. (2018). [Volatility is rough](#). *Quantitative Finance*, 18(6), 933–949.
- ▣ Glasserman, P. (2003). [Monte Carlo Methods in Financial Engineering](#). Springer.
- ▣ Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (Eds.) (2017). [Applied Quantitative Finance](#), 3rd ed. Springer.
- ▣ Heston, S.L. (1993). [A closed-form solution for options with stochastic volatility with applications to bond and currency options](#). *Review of Financial Studies*, 6(2), 327–343.
- ▣ Higham, D.J. (2001). [An algorithmic introduction to numerical simulation of stochastic differential equations](#). *SIAM Review*, 43(3), 525–546.
- ▣ Itô, K. (1944). [Stochastic integral](#). *Proceedings of the Imperial Academy*, 20(8), 519–524.
- ▣ Kloeden, P.E., Platen, E. (1992). [Numerical Solution of Stochastic Differential Equations](#). Springer.
- ▣ Lee, S.S., Mykland, P.A. (2008). [Jumps in financial markets: a new nonparametric test and jump dynamics](#). *Review of Financial Studies*, 21(6), 2535–2563.
- ▣ Lord, R., Koekoek, R., van Dijk, D. (2010). [A comparison of biased simulation schemes for stochastic volatility models](#). *Quantitative Finance*, 10(2), 177–194.
- ▣ Merton, R.C. (1973). [Theory of rational option pricing](#). *Bell Journal of Economics and Management Science*, 4(1), 141–183.

References (3/3)

- ▣ Merton, R.C. (1976). Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics*, 3(1–2), 125–144.
- ▣ Milstein, G.N. (1975). Approximate integration of stochastic differential equations. *Theory of Probability and Its Applications*, 19(3), 557–562.
- ▣ Nelson, D.B. (1990). ARCH models as diffusion approximations. *Journal of Econometrics*, 45(1–2), 7–38.
- ▣ Osborne, M.F.M. (1959). Brownian motion in the stock market. *Operations Research*, 7(2), 145–173.
- ▣ Pele, D.T. (2014). A SAS approach for estimating the parameters of an alpha-stable distribution. *Procedia Economics and Finance*, 10, 68–77.
- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2019). Using high-frequency entropy to forecast Bitcoin's daily value at risk. *Entropy*, 21(2), 102.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets?. *European Journal of Finance*, 29(10), 1064–1105.
- ▣ Samuelson, P.A. (1965). Rational theory of warrant pricing. *Industrial Management Review*, 6(2), 13–39; reprinted in *Henry P. McKean Jr. Selecta* (2015), Birkhäuser.
- ▣ Tang, C.Y., Chen, S.X. (2009). Parameter estimation and bias correction for diffusion processes. *Journal of Econometrics*, 149(1), 65–81.
- ▣ Uhlenbeck, G.E., Ornstein, L.S. (1930). On the theory of the Brownian motion. *Physical Review*, 36(5), 823–841.
- ▣ Vasicek, O. (1977). An equilibrium characterization of the term structure. *Journal of Financial Economics*, 5(2), 177–188.
- ▣ Wiener, N. (1923). Differential-space. *Journal of Mathematics and Physics*, 2(1–4), 131–174.